



Introduction to Cryptography

Lab 11. AES-CTR

May 28, 2026

1. Learning to use AES

Do the following exercises by your own. Choose a cryptographic library (in C/C++, Python, Java, C#) to develop the following exercises. Complete all the exercises in this section and upload to Microsoft Teams before the session ends: screenshots of your source code and screenshots of your program running showing the results obtained. **If you do all the exercises in this section during the session you will get 2 points**

1. Implement a program to generate a random key for AES. Try to generate keys of 128, 192 and 256 bits. Print the generated key in base64.
2. Implement a program to encipher a string using AES and counter mode. Your program must randomly generate an IV. Print the IV used to encipher and the ciphertext in base64.
3. Implement a program to decipher using AES and counter mode. It must decipher a ciphertext encoded in base64. Print the recovered plaintext.

2. Using AES properly

Do the following exercises in pairs. Choose a cryptographic library (in C/C++, Python, Java, C#) to develop the following exercises.

1. Design and implement a program to generate a random key for AES. Your program must fulfill the following requirements:
 - The key size must be given by the user.

- The generated key must be stored in a file, encoded in base64. The filename must be given by the user.
2. Design and implement a program to encipher a plaintext using AES and the counter mode. Your program must fulfill the following requirements:
 - The user must give the filename where the key is stored.
 - The user must give the filename of the plaintext. The plaintext could be a file of any kind: a pdf, source code, a docx, etc.
 - Your program must be able to encipher files of any size (Kb, Mb, Gb, etc,)
 - The counter or nonce must be used only once and must be generated at random.
 - The output, i.e. the ciphertext must be stored in a file encoded in base64. The filename of the ciphertext must be given by the user.
 3. Design and implement a program to decipher a plaintext using AES and the counter mode. Your program must fulfill the following requirements:
 - The user must give the filename where the key is stored.
 - The user must give the filename of the ciphertext.
 - The recovered plaintext must be stored in a file.
 4. You must show your programs in class, as follows:
 - One student must generate a random key and encipher a file. Then he/she will share with a partner the key and the file of the ciphertext.
 - Once the partner received the key and the ciphertext, he/she must be able to decipher it and recover the plaintext.

3. Questions

Answer the following questions:

1. Which programming language did you use? Which cryptographic library did you use?
2. What is the name of the function you use to generate an AES key? How many parameters does it require? Describe in your own words, what is each parameter?

3. What is the name of the function or the instruction did you use to generate the counter or nonce? What is the size of this data? Can you vary the size of counter mode or nonce? How?
4. What is the name of the function did you use to encipher? How many parameters does it require? Describe in your own words, what is each parameter?
5. If you want to use a different mode of operation, e.g. CBC, GCM, OFB, etc, what function or instruction you must change?
6. What is the name of the function did you use to decipher? How many parameters does it require? Describe in your own words, what is each parameter?

Products that you must upload to Teams

Every student must write a report to present the results. Your document must include

- Full name of the student, date of the lab session and the topic we are studying.
- Answers to questions of Section 3.